Task 8: Project Documentation

Introduction

This project demonstrates the deployment of a secure cloud-based web server on Amazon Web Services (AWS). The goal was to design a cloud network, deploy an EC2 instance, configure secure access using Security Groups, install and configure Nginx, and host a custom web page accessible from the internet. The project also provided hands-on experience with AWS networking components and Linux server administration.

Architecture

The solution consists of the following components:

- VPC (10.0.0.0/16)
- Public Subnet (10.0.1.0/24)
- Internet Gateway
- Route Table
- Security Group
- EC2 Instance
- Nginx Web Server
- Internet Users

Deployment Steps

Step 1: Create a VPC

Created a custom VPC using the CIDR block:

    10.0.0.0/16

Step 2: Create a Public Subnet

Created a subnet with:

    10.0.1.0/24

Step 3: Configure Internet Connectivity

- Created an Internet Gateway.
- Attached the Internet Gateway to the VPC.
- Created a Route Table.
- Added a default route:

0.0.0.0/0 → Internet Gateway

- Associate the Route Table with the public subnet.

Step 4: Launch EC2 Instance

- Ubuntu Server
- Public IP enabled
- Deployed inside the public subnet

Step 5: Configure Security Groups

Allowed:

Port	Protocol	Purpose
22	TCP	SSH
80	TCP	HTTP
443	TCP	HTTPS

Step 6: Install and Configure Nginx

Connected via SSH and executed:

```
sudo apt update
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

Created a custom HTML page and deployed it to:

```
/var/www/html/index.html
```

Security Measures

1. **Security Groups**

Security Groups were configured as a virtual firewall to control inbound traffic.

Rules configured:

- SSH (22) is restricted to my IP address only.

- HTTP (80) open to the internet.
- HTTPS (443) open to the internet.

This reduced unnecessary exposure to external threats.

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2. **IAM User**

An IAM user was used instead of the AWS root account for managing AWS resources.

Benefits:

- Principle of least privilege.
- Improved security.
- Better accountability through user-specific actions.

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3. **SSH Restrictions**

SSH access was restricted to a single trusted public IP address rather than allowing access from anywhere.

Example:

My-IP/32

This significantly reduced the attack surface of the EC2 instance.

Challenges and Solutions

Challenge	Cause	Solution
Unable to SSH into EC2	Security Group blocked port 22	Added SSH rule for my IP
Website inaccessible	Port 80 not allowed	Added HTTP rule
Nginx not serving page	Service not running	Started and enabled Nginx

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Lessons Learned

This project strengthened my understanding of:

- AWS VPC architecture
- CIDR block planning
- Public and private networking concepts
- Route Tables and Internet Gateways
- Security Group configuration
- EC2 deployment and management
- Linux administration using SSH
- Installing and managing Nginx
- Basic cloud troubleshooting techniques
- Implementing security best practices in AWS

The project provided practical experience in deploying and securing cloud infrastructure, which is a fundamental skill for cloud and DevOps engineers.